Unit 7 Guided Notes — Data Analysis & Statistics

ANSWER KEY

Scatterplots, correlation, the line of best fit, and two-way tables.

🎯 By the end of this unit, I can…

- Describe association in a scatter plot
- Draw and use a line of best fit to make predictions
- Read and build two-way tables and relative frequencies

Words to know

- **Bivariate data** — two numbers for each person or thing.
- **Scatter plot** — a graph that shows each pair of numbers as one dot.
- **Association** — two things change together in a pattern.
- **Outlier** — a dot far away from the rest.
- **Line of best fit** — one straight line through the middle of the dots.
- **Two-way table** — a table that counts people two ways at once.
- **Relative frequency** — a count divided by a total (a fraction or percent).
- **Lurking variable** — a hidden third thing that makes two things change together.

Standards: NC.8.SP.1 · NC.8.SP.2 · NC.8.SP.3 · NC.8.SP.4

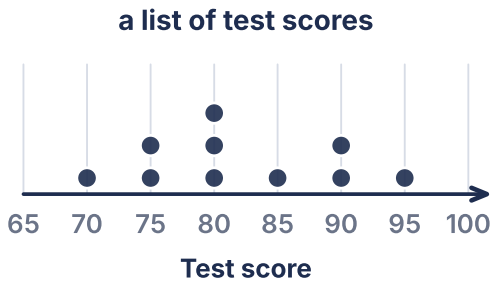
Answer key: every blank and final answer is filled in green.

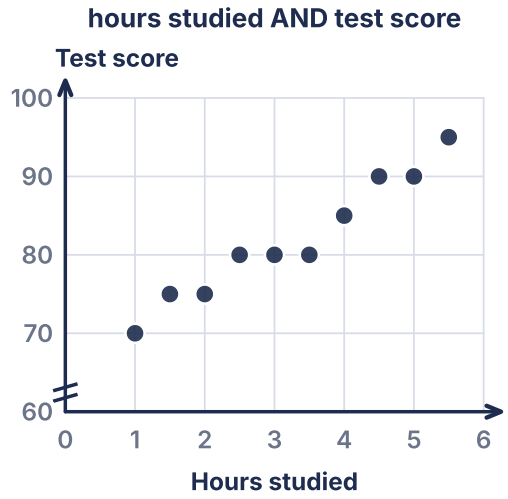
1 Scatterplots & Bivariate Data

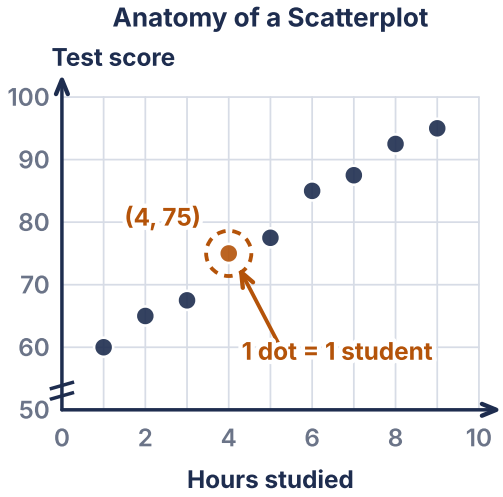
📌 **NC.8.SP.1:** Construct and interpret scatter plots for bivariate measurement data.
I can make a scatter plot and tell what it shows.

Bivariate data has TWO values for each item — like a person's height AND weight, or hours studied AND test score. We use a scatterplot to look for patterns between the two variables.

Term	Meaning	Example
Univariate data	ONE variable per data point	a list of test scores
Bivariate data	TWO variables per data point	hours studied AND test score





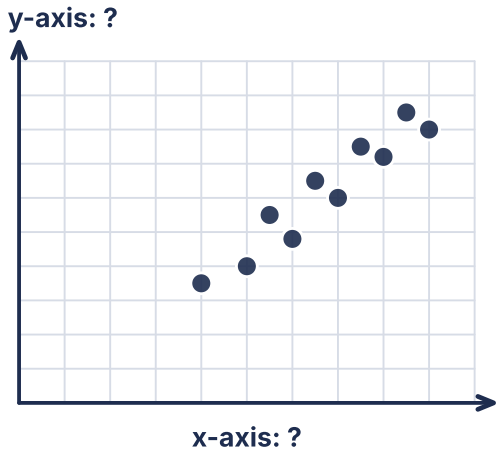


A scatterplot has: an x-axis for the **independent (input)** variable, a y-axis for the **dependent (output)** variable. Each dot represents **one** data point. Both axes need a **label** and a **scale**.

📊 **The two variables together form an ordered pair (x, y) that we plot as a single dot.** Each dot is one person, one observation.

Try It

Suppose you collect data on 10 students: how many hours of sleep they got and their test score the next day.



Which data goes across? Which goes up? Count the dots.

- Independent variable (x-axis)? **Hours of sleep**
- Dependent variable (y-axis)? **Test score**
- Total number of dots on the scatterplot: **10**

💡 **Why scatterplots?** Bar graphs count things in groups. Scatter plots show if two numbers change together. If you're asking "does X affect Y?", you want a scatterplot.

✏ Practice: Scatterplots

1. ★ Try it Which data set is **bivariate**?

Heights

Student	Height (cm)
Ana	152
Ben	160
Cal	148

Sleep and test score

Student	Sleep (h)	Score
Ana	8	85
Ben	6	72
Cal	9	90

Favorite colors

Student	Color
Ana	blue
Ben	red
Cal	green

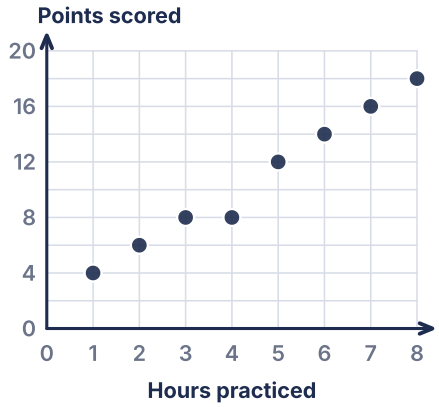
- ○ Heights of 20 students
- ● **Hours of sleep AND test score for 20 students**
- ○ Favorite colors of 20 students

How to do it: Bivariate = two values for each item. Hours of sleep AND test score gives each student two numbers, so it is bivariate. The other two have only one value per student.

2. ★★ On level The scatterplot shows hours of practice and points scored for 8 players. How many points did the player who practiced **5 hours** score?

Answer: 12

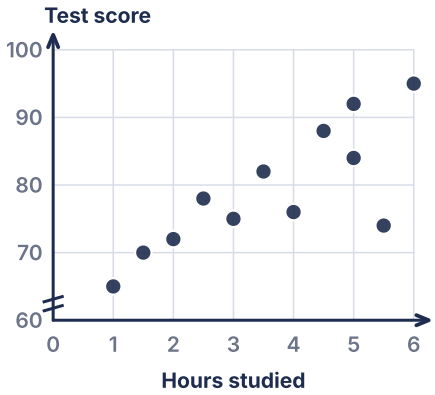
How to do it: Find 5 on the x-axis. Go up to the dot. Go left: the dot is on the line for 12. The player scored 12 points.



3. ★★★ Challenge How many students studied **more than 4 hours** AND scored **above 80**?

Answer: 4

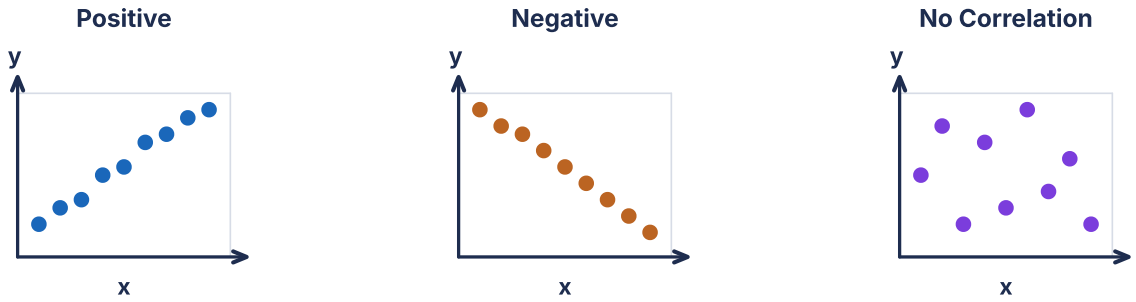
How to do it: Dots to the right of 4 hours: 4.5 h (88), 5 h (84), 5 h (92), 5.5 h (74), 6 h (95). Above 80: 88, 84, 92, 95. 74 is below 80. So 4 students.



2 Correlation

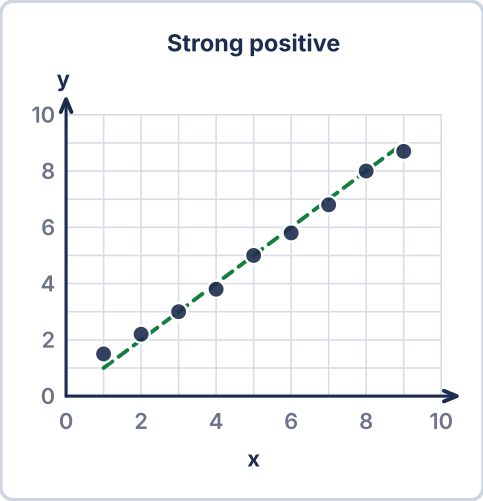
📌 **NC.8.SP.1:** Describe patterns such as clustering, outliers, positive or negative association, linear association, and nonlinear association.
I can tell if the dots go up, go down, or show no pattern, and spot outliers.

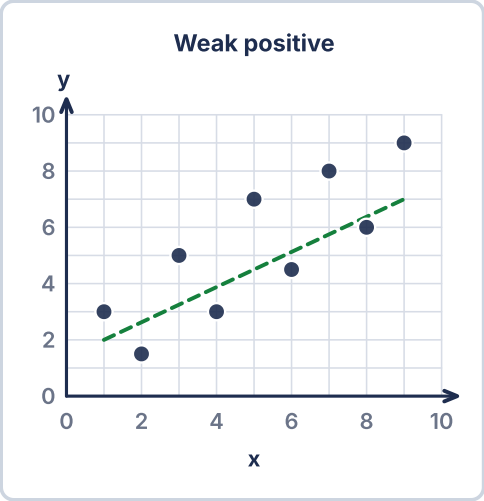
Correlation describes how two variables tend to change together — **positive**, **negative**, or **none**. Plus two strengths: **strong** or **weak**.



Positive correlation: as x increases, y also increases. Negative correlation: as x increases, y decreases. No correlation: the dots are scattered with no clear pattern.

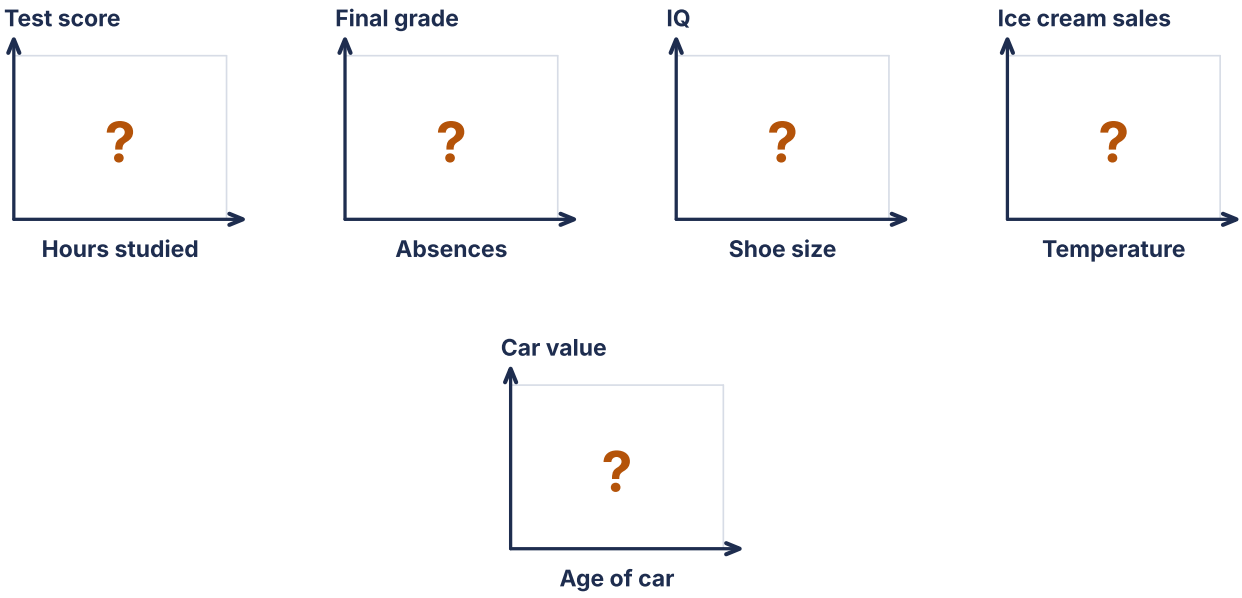
Strength of Correlation





Strong = dots tightly hug the trend line. Weak = dots are spread out around the trend.

Try It — Predict the correlation type

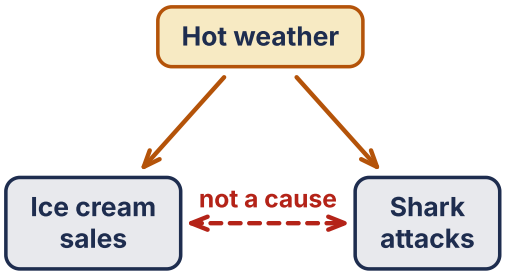


Picture the dots on each graph before you answer.

- Hours studied vs. test score → **Positive**
- Number of absences vs. final grade → **Negative**
- Shoe size vs. IQ → **None**
- Outdoor temperature vs. ice cream sales → **Positive**
- Age of a car vs. value of the car → **Negative**

🎯 **Outliers and clusters:** An *outlier* is a dot that's far from the rest. A *cluster* is a group of dots tightly grouped together. Both are worth pointing out when you describe a scatterplot.

⚠️ **CORRELATION ≠ CAUSATION:** Just because two things are correlated doesn't mean one causes the other! Ice cream sales and shark attacks are both higher in summer — but ice cream doesn't cause sharks. Both are caused by hot weather.

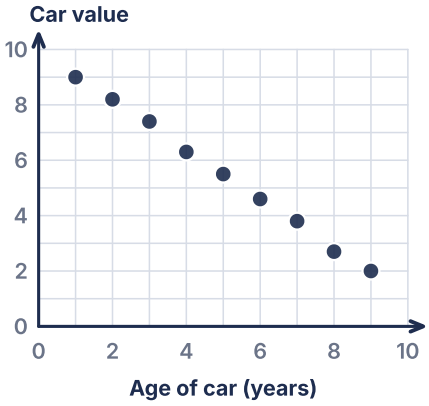


✏ Practice: Correlation

1. ★ Try it What type of correlation does this scatterplot show?

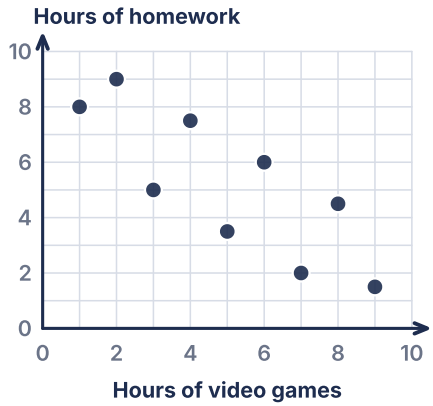
- ○ Positive
- ● **Negative**
- ○ None

How to do it: From left to right the dots go DOWN. As x increases, y decreases. That is a negative correlation.



2. ★★ On level Describe the correlation. Pick the direction AND the strength.

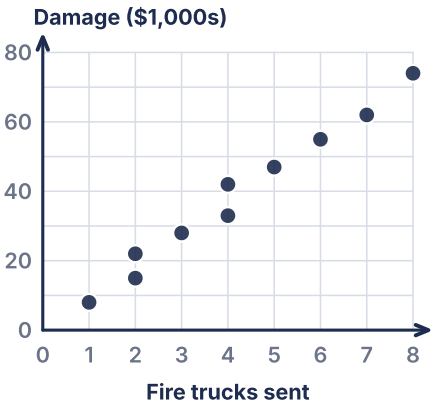
- ○ Strong positive
- ○ Weak positive
- ○ Strong negative
- ● **Weak negative**



> **How to do it:** Direction: the dots drift DOWN from left to right → negative. Strength: the dots are spread out, not hugging a line → weak. Answer: weak negative.

3. ★★★ Challenge Fires where more fire trucks show up have more damage. Does sending more trucks **cause** more damage?

- ○ Yes — more trucks cause more damage
- ● **No — big fires bring more trucks AND cause more damage**
- ○ No — there is no correlation at all



> **How to do it:** There IS a positive correlation, but it is not cause and effect. The **lurking variable** is the size of the fire: a big fire brings more trucks AND does more damage.

3 Line of Best Fit

📌 **NC.8.SP.2 / NC.8.SP.3:** Informally fit a straight line for scatter plots that suggest a linear association; use the equation to solve problems and make predictions.
I can draw a line that fits the dots and use it to predict.

When the dots suggest a linear pattern, we can draw a single straight line through the "middle" of the cloud of dots. This line represents the overall trend and helps us make predictions.

When Should You Draw One?

A line of best fit only makes sense when:

- The data shows a **linear** pattern (positive or negative correlation).
- You should NOT draw one if the dots show **no** correlation or a curved/non-linear pattern.

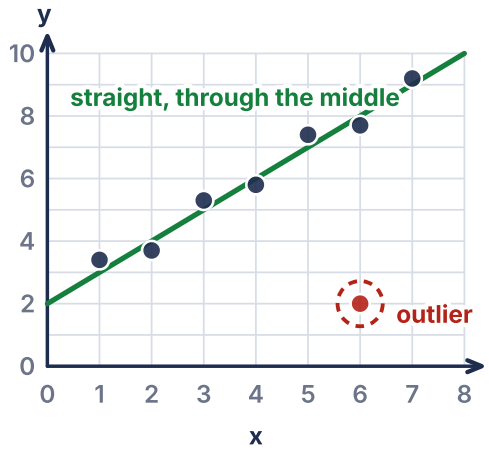


4 dots above, **4 dots below**, 1 right on the line.

Notice the line: passes through the "middle" of the dots · has roughly the same number of dots above and below it · follows the overall positive trend · doesn't have to touch any specific dot.

Drawing a Line of Best Fit (Informally)

1. Look at the overall direction of the dots (positive or negative).
2. Use a ruler — put the line so dots are roughly balanced above and below.
3. Don't connect the dots! It's a smooth straight line, not a zigzag.
4. Don't be tricked by **outliers** — ignore dots far from the rest.



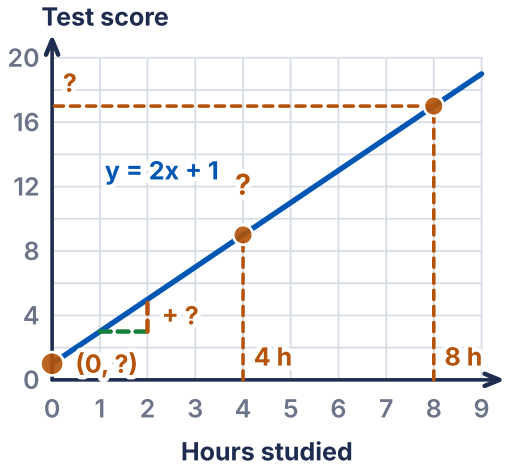
One ruler line. Dots balanced above and below. The red outlier is ignored.

Using the Line to Make Predictions

Once you draw a line of best fit, you can:

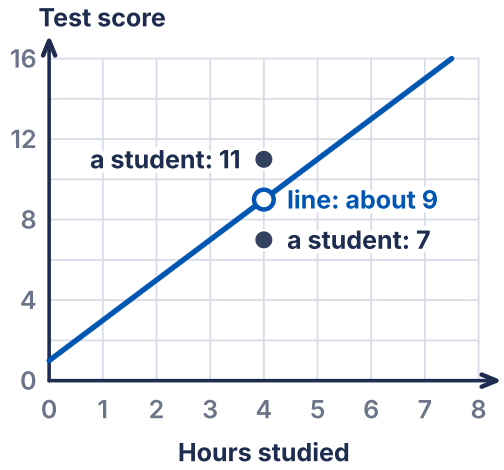
- Find its equation (**y = mx + b**) by reading off the slope and y-intercept.
- Plug in a new x-value to predict y.
- Read graphically: find x on the x-axis, go up to the line, then over to the y-axis.

Try It — y = 2x + 1 (x = hours studied, y = test score)



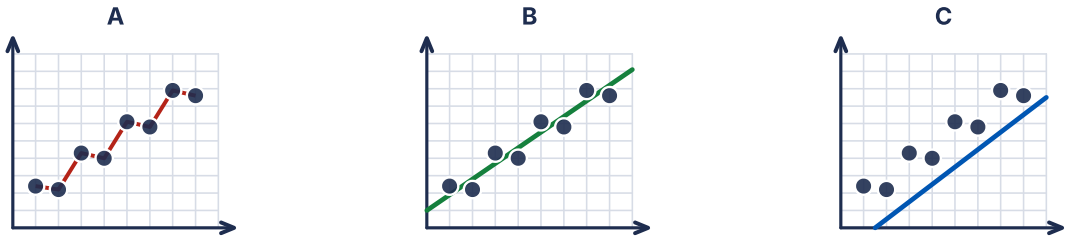
- The slope means: each extra hour of study adds about 2 points.
- The y-intercept means: with 0 hours of study, the predicted score is 1.
- If a student studies 4 hours, predicted score = 2(4) + 1 = 9.
- If a student studies 8 hours, predicted score = 17.

⚠️ **Predictions are approximate.** The line is a model, not exact. A student who studies 4 hours might get 7 or 11 — the line predicts about 9 on average.



✏ Practice: Line of Best Fit

1. ★ Try it Which graph shows a good line of best fit?



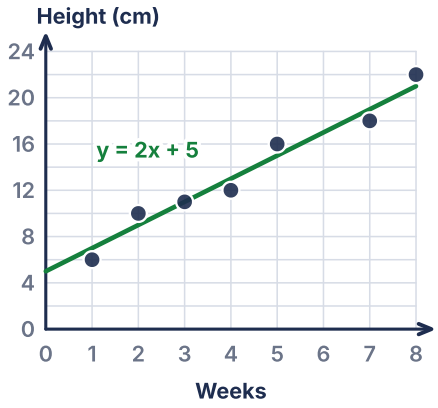
- ○ Graph A
- ● **Graph B**
- ○ Graph C

> **How to do it:** A connects the dots (a zigzag, not a line). C has every dot above the line. B is one straight line with 4 dots above and 4 below. Graph B.

2. ★★ On level The line of best fit for a plant is y = 2x + 5. x = weeks and y = height in cm. Predict the height at **6 weeks**.

Answer: 17

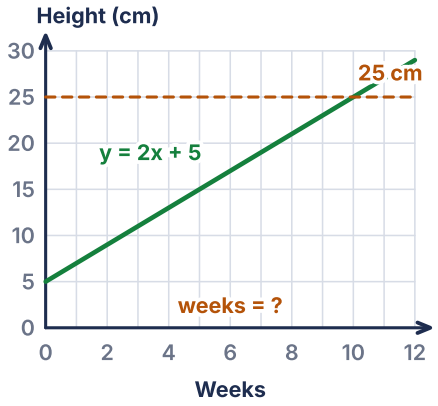
How to do it: y = 2x + 5 → y = 2(6) + 5 = 12 + 5 = 17. The plant should be about 17 cm tall.



3. ★★★ Challenge Use the same line, y = 2x + 5. About how many **weeks** until the plant is **25 cm** tall?

Answer: 10

How to do it: Set y = 25: 2x + 5 = 25. Subtract 5: 2x = 20. Divide by 2: x = 10. About 10 weeks.



4 Two-Way Tables

📌 **NC.8.SP.4:** Understand patterns of association between two variables by displaying frequencies and relative frequencies in a two-way table.
I can read a two-way table and find percents in it.

A **two-way table** counts people two ways at once. Example: grade level AND whether they like pizza. Both things are groups (categories), not numbers.

When do we use them? Two-way tables help organize *categorical* bivariate data — when both variables are categories instead of numbers. Example: grade level and lunch preference, gender and favorite sport.

Reading a Two-Way Table — 120 students surveyed about pizza

	Like Pizza	Don't Like Pizza	Total
6th Grade	32	8	40
7th Grade	28	12	40
8th Grade	24	16	40
Total	**84**	**36**	**120**

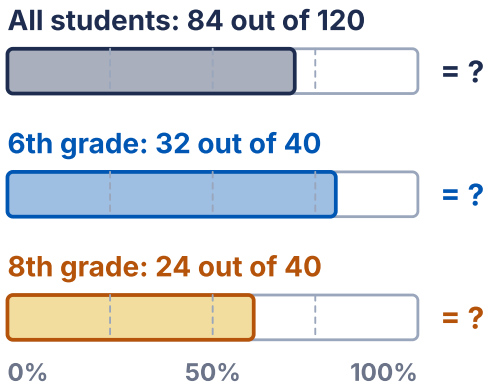
	Like pizza	Do not like	Total
6th Grade	32	8	40
7th Grade	28	12	40
8th Grade	24	16	40
Total	84	36	120

A row and a column cross at one cell: 6th grade AND likes pizza. Totals are on the edges.

- Total students surveyed: 120
- Number of 6th graders who like pizza: 32
- Total 7th graders surveyed: 40
- Total students who don't like pizza: 36

Relative Frequency

Relative frequency = the count divided by a total. Same idea as in Unit 6 frequency tables, but now we have multiple totals to choose from.



- Fraction of all 120 students who like pizza: 84 ÷ 120 = 0.7 or 70%
- Fraction of 6th graders who like pizza: 32 ÷ 40 = 0.8 or 80%
- Fraction of 8th graders who like pizza: 24 ÷ 40 = 0.6 or 60%

💡 **Looking for an association:** 80% of 6th graders like pizza but only 60% of 8th graders do. There's an **association** between grade level and pizza preference — they're related!

Try It — 100 students surveyed about pets

	Has pet	No pet	Total
Boys	20	30	50
Girls	35	15	50
Total	**55**	**45**	**100**

	Has pet	No pet	Total
Boys	20	30	50
Girls	35	15	50
Total	55	45	100

Boys: $\frac{20}{50}$ = ?

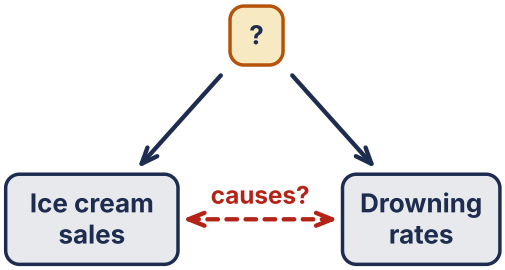
Girls: $\frac{35}{50}$ = ?

- Fraction of boys who have a pet: 20 ÷ 50 = **0.4** or **40%**
- Fraction of girls who have a pet: 35 ÷ 50 = **0.7** or **70%**
- Is there an association between gender and pet ownership?

  Yes — girls are much more likely to have one

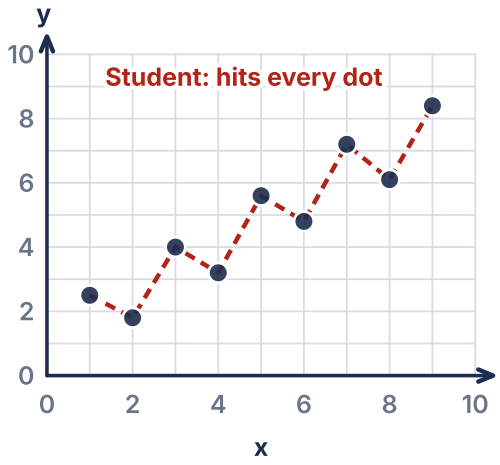
Error Analysis

SPOT THE ERROR **"Ice cream sales and drowning rates have a positive correlation, so eating ice cream causes drowning."**



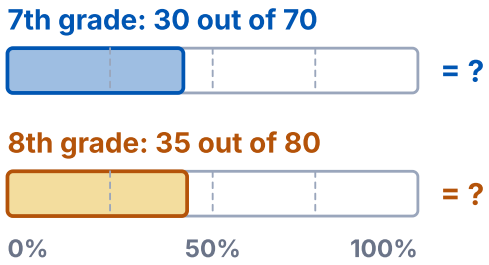
Correlation does NOT mean causation! Both increase in summer because of a lurking variable — hot weather. Always look for a third variable that could explain the relationship.

SPOT THE ERROR **"The line of best fit must pass through every data point."**



The line of best fit shows the general trend — it should have roughly equal points above and below, but does NOT need to pass through any specific point.

SPOT THE ERROR "More 8th graders like basketball (35) than 7th graders (30), so 8th graders are more into basketball." (8th grade has 80 students total vs 70 in 7th grade!)



Use relative frequency to compare fairly: 7th = $^{30}/_{70}$ ≈ 42.9% vs 8th = $^{35}/_{80}$ ≈ 43.75%. They're nearly the same! When comparing groups of different sizes, ALWAYS use percentages, not raw counts.

✏ Practice: Two-Way Tables

1. ★ Try it 80 students were asked if they play a sport. How many **8th graders play a sport**?

Answer: 27

How to do it: Row: 8th Grade. Column: Plays a sport. The cell where they cross is 27.

	Plays a sport	No sport	Total
7th Grade	18	22	40
8th Grade	27	13	40
Total	**45**	**35**	**80**

2. ★★ On level Same table. What **percent of 8th graders** play a sport? Write it as a percent.

Answer: 67.5%

How to do it: 8th graders who play ÷ all 8th graders = 27 ÷ 40 = 0.675. × 100 → 67.5%.

	Plays a sport	No sport	Total
7th Grade	18	22	40
8th Grade	27	13	40
Total	**45**	**35**	**80**

3. ★★★ Challenge Same table. Is there an **association** between grade and playing a sport?

	Plays a sport	No sport	Total
7th Grade	18	22	40
8th Grade	27	13	40
Total	**45**	**35**	**80**

● **Yes — 67.5% of 8th graders play, but only 45% of 7th graders do**

○ No — both grades have 40 students

○ No — the two grades have about the same percent

How to do it: 7th: 18 ÷ 40 = 0.45 = 45%. 8th: 27 ÷ 40 = 0.675 = 67.5%. The percents are far apart, so yes — grade and playing a sport are associated.

Practice — All Sections

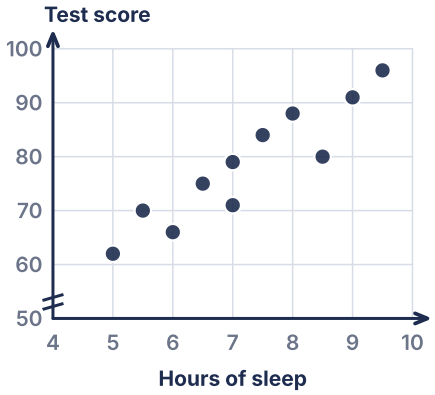
Mixed review of scatterplots, correlation, the line of best fit, and two-way tables. Look back at the notes if you get stuck.

✏ Comprehensive Practice All sections

1. Score 3 Each dot is one student. How many students are shown?

Answer: 11

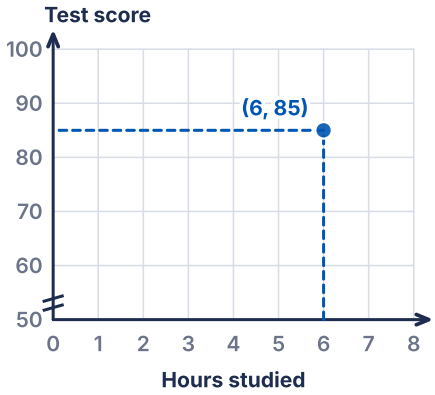
How to do it: Count every dot one at a time: there are 11 dots, so 11 students.



2. Score 3 On a scatterplot, x = hours studied and y = test score. What does the dot **(6, 85)** mean?

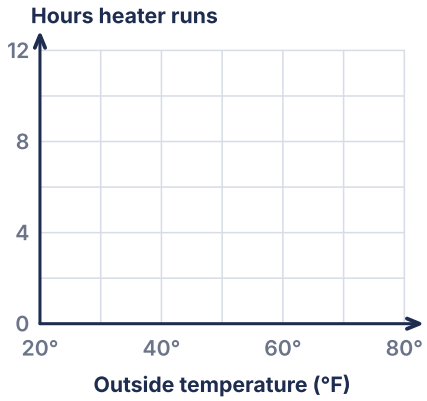
- ○ A student studied 85 hours and scored 6
- ● **A student studied 6 hours and scored 85**
- ○ 6 students scored 85

How to do it: (x, y) = (6, 85). x = 6 hours studied, y = a score of 85. One dot = one student.

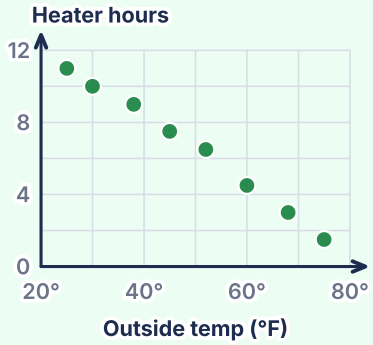


3. Score 3 Outside temperature vs. hours the heater runs. What type of correlation would you expect? Picture the dots on these axes.

- ○ Positive
- ● **Negative**
- ○ None



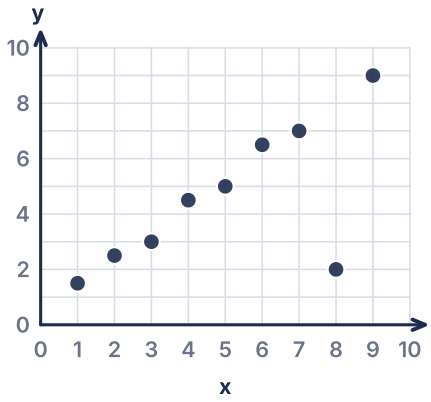
How to do it: As the temperature goes UP, the heater runs LESS. One goes up while the other goes down → negative.



4. Score 4 One dot is an **outlier**. Write it as an ordered pair (x, y).

Answer: (8, 2)

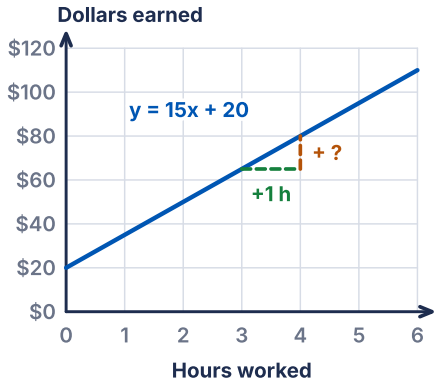
How to do it: The dots go up in a tight line, but the dot at x = 8 is way down at y = 2. The outlier is (8, 2).



5. Score 4 A line of best fit for Mia's pay is y = 15x + 20. x = hours worked and y = dollars earned. What does the **15** mean?

- ● **She earns $15 more for each extra hour**
- ○ She starts with $15
- ○ She works 15 hours

How to do it: 15 is the slope (the number times x). Each extra hour (x + 1) adds about $15. The 20 is the y-intercept: the starting amount.



6. Score 4 Find the missing number in the table.

Answer: 18

How to do it: The 6th grade row adds to 30: 12 + ? = 30, so ? = 30 − 12 = 18. Check the column: 18 + 9 = 27 ✓.

	Has a phone	No phone	Total
6th Grade	12	?	30
7th Grade	21	9	30
Total	**33**	**27**	**60**

7. Score 4 What **percent of girls** prefer cats? Write it as a percent.

Answer: 70%

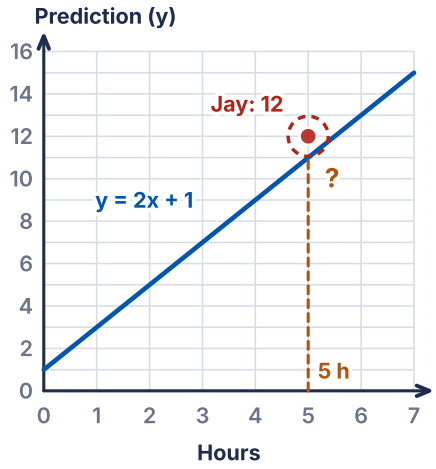
How to do it: 35 girls prefer cats out of 50 girls: 35 ÷ 50 = 0.7 → 70%.

	Dogs	Cats	Total
Boys	24	16	40
Girls	15	35	50
Total	**39**	**51**	**90**

8. Score 5 Jay used y = 2x + 1 for 5 hours and wrote 2 × 5 + 1 = 2 × 6 = 12. What is the correct prediction?

Answer: 11

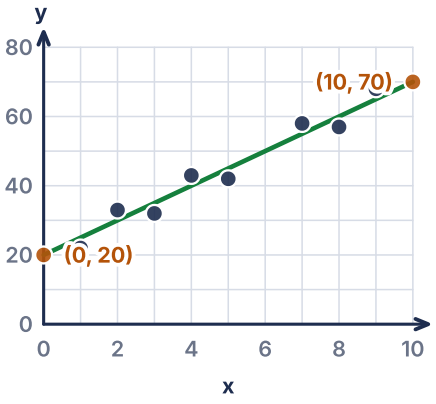
How to do it: Jay added 5 + 1 first. Multiply first: y = 2(5) + 1 = 10 + 1 = 11. The correct prediction is 11.



9. Score 5 The line of best fit goes through (0, 20) and (10, 70). Predict y when **x = 6**.

Answer: 50

How to do it: Slope = (70 − 20) ÷ (10 − 0) = 50 ÷ 10 = 5. The y-intercept is 20, so y = 5x + 20. At x = 6: y = 5(6) + 20 = 30 + 20 = 50.



10. Score 5 Same table. Boys are more likely to prefer dogs. By how many **percentage points**?

Answer: 30

How to do it: Boys: 24 ÷ 40 = 0.6 = 60%. Girls: 15 ÷ 50 = 0.3 = 30%. 60% − 30% = 30 percentage points. (Use percents, not counts — the groups are different sizes.)

	Dogs	Cats	Total
Boys	24	16	40
Girls	15	35	50
Total	**39**	**51**	**90**